AI 1100 - Calculus and Linear Algebra

Instructor

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Week 8 Functions and limits
Subtopics: Functions, limit, continuity

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OF ARTIFICIAL INTELLIGENCE

Course Learning Outcome

1. Solve systems of linear equations using Gaussian eliminations, matrix factorizations. Explain core properties of regular and singular matrices and their applications.
2. Explain and apply the main concepts of vector spaces and linear transformations, including basis, dimension, orthogonality, and matrix representations.
3. Explain main concepts of real-valued functions such as continuity, differentiability and their applications in modeling real world problems.
4. Explain the main properties of integrals and their applications.
5. Use basic linear algebra and calculus tools to model and analyze applied problems.
6. Develop critical thinking towards real world problems. Develop the ability to pass from concrete problems to their mathematical abstractions and recognize necessary tools from Linear Algebra and Calculus in order to solve them.

Functions

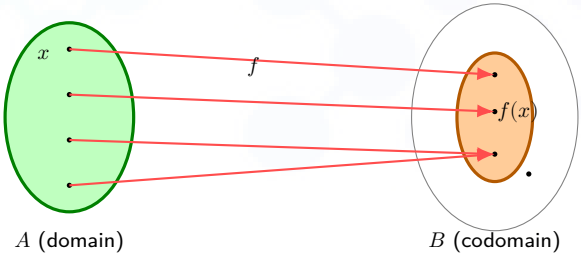
Definition 1 (Function)

A function f is a rule that assigns each element x in a set A to exactly one element, called $f(x)$ in a set B. The set A is called the domain of f, and the set B is called the codomain of f.

Remark: Let A and B be a set, and $f : A \to B$ be a function. We have further terminologies

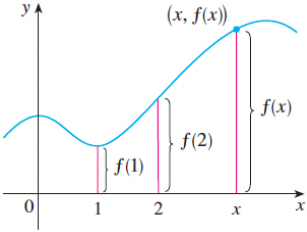
- **Natural domain**: the largest set on which the formula for f is defined.
- **Range**: the set of outputs actually attained by f.

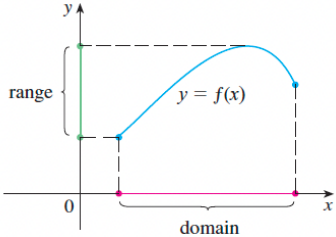
$$\mathrm{Range}(f) = \{\, f(x) : x \in A \,\} = \{\, y \in B : \ \exists\, x \in A,\ y = f(x) \,\}.$$



 Example. For $f(x) = x^2$ natural domain $= \mathbb{R}$, $\quad \mathrm{Range}(f) = [0, \infty)$.

Illustration





Idea

- The **domain** is the set of inputs x for which the formula makes sense (real output).
- The **range** is the set of all possible outputs $y = f(x)$.

To determine the domain we watch for

- division by 0
- even roots of negatives
- logs of nonpositives

Example: Finding the domain

Example 1. Find the natural domain of the following functions

(a) $g(x) = \dfrac{x+3}{x^2+3x-10}$ **(b)** $f(x) = \sqrt{5-3x}$

(c) $h(x) = \dfrac{\sqrt{7x+8}}{x^2+4}$ **(d)** $R(x) = \dfrac{\sqrt{10x-5}}{x^2-16}$

Solution.

(a) $g(x) = \dfrac{x+3}{x^2+3x-10}$

$$x^2+3x-10 = (x+5)(x-2) \neq 0$$

$\therefore$ Domain : $D_g = \mathbb{R} \setminus \{-5, 2\}$.

(b) $f(x) = \sqrt{5-3x}$

$$5-3x \geq 0$$
$$x \leq \tfrac{5}{3}$$

$\therefore$ Domain : $D_f = (-\infty, \tfrac{5}{3}]$.

(c) $h(x) = \dfrac{\sqrt{7x+8}}{x^2+4}$

$$7x+8 \geq 0 \Rightarrow x \geq -\tfrac{8}{7},$$
$$x^2+4 > 0 \text{ for all } x,$$

$\therefore$ Domain $=: D_h[-\tfrac{8}{7}, \infty)$.

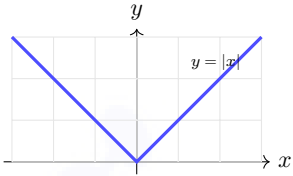
(d) $R(x) = \dfrac{\sqrt{10x-5}}{x^2-16}$

$$10x-5 \geq 0 \Longrightarrow x \geq \tfrac{1}{2},$$
$$x^2-16 \neq 0 \Longrightarrow x \neq \pm 4,$$

$\therefore$ Domain : $D_R = [\tfrac{1}{2}, \infty) \setminus \{4\}$.

Absolute value

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0. \end{cases}$$



Quick checks

$$|-3| = -(-3) = 3, \qquad |2| = 2, \qquad |t| = \begin{cases} t, & t \geq 0, \\ -t, & t < 0. \end{cases}$$

Piecewise function

A function defined by *different formulas* on *different parts* of the domain.

$$h(x) = \begin{cases} h_1(x), & \text{on region 1}, \\ h_2(x), & \text{on region 2}, \\ \vdots \end{cases}$$

How to evaluate

1. Decide in which region your input satisfies.
2. Use *only* the corresponding formula.

Evaluate a Piecewise Function

Example 2. Given

$$g(t) = \begin{cases} 3t^2 + 4 & \text{if } t \leq -4, \\ 10 & \text{if } -4 < t \leq 15, \\ 1 - 6t & \text{if } t > 15. \end{cases}$$

Evaluate :

(a) $g(-6)$, **(b)** $g(-4)$, **(c)** $g(1)$, **(d)** $g(15)$, **(e)** $g(21)$

Solution.

(a) Since $-6 \leq -4$, to evaluate $g(-6)$ we use top piece

$$\begin{aligned} g(-6) &= 3(-6)^2 + 4 \\ &= 3 \cdot 36 + 4 \\ &= \boxed{112} \end{aligned}$$

(b) To evaluate $g(-4)$, since -4 lies in the boundary in top piece, then

$$\begin{aligned} g(-4) &= 3(-4)^2 + 4 \\ &= 3 \cdot 16 + 4 \end{aligned}$$

(c) $g(1)$ ($t = 1 \in (-4, 15]$ (middle))

$$g(1) = 10 = \boxed{10}$$

(d) $g(15)$ (boundary in middle piece)

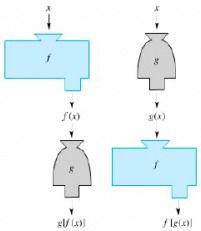
$$g(15) = 10 = \boxed{10}$$

(e) $g(21)$ (bottom piece)

$$\begin{aligned} g(21) &= 1 - 6(21) \\ &= 1 - 126 = \boxed{-125} \end{aligned}$$

Function Composition

Function Composition



$g \circ f$ $\qquad$ $f \circ g$

Let f and g be functions. The composition of g with f is

$$(g \circ f)(x) = g(f(x)), \qquad (f \circ g)(x) = f(g(x)).$$

In general, composition is **not** commutative:

$$(g \circ f)(x) \neq (f \circ g)(x).$$

Domain of $(g \circ f)$:

$$D_{g \circ f} = \{\, x : \ x \in \mathrm{Dom}(f) \text{ and } f(x) \in \mathrm{Dom}(g) \,\}.$$

Example 3. Let $f(x) = x + 3$ and $g(x) = \sqrt{x}$. Calculate $(f \circ g)(x)$ and $(g \circ f)(x)$.

$$(f \circ g)(x) = f(g(x)) = f(\sqrt{x}) = \sqrt{x} + 3,$$
$$(g \circ f)(x) = g(f(x)) = g(x + 3) = \sqrt{x + 3}.$$

Hence we see that

$$\sqrt{x + 3} \neq \sqrt{x} + 3.$$
$$(g \circ f)(x) \neq (f \circ g)(x).$$

Example: Evaluating Composition

Example 4. Let $f(x) = \dfrac{6x}{x^2 - 9}$ and $g(x) = \sqrt{3x}$

(a) Evaluate $(f \circ g)(12)$.

(b) Find $(f \circ g)(x)$ and its domain.

Solution.

(a) Evaluate $(f \circ g)(12)$.

$$(f \circ g)(12) = f(g(12)) = f(\sqrt{36}) = f(6) = \frac{6 \cdot 6}{6^2 - 9} = \frac{36}{27} = \frac{4}{3}.$$

(b) Find $(f \circ g)(x)$ and its domain.

$$(f \circ g)(x) = f(\sqrt{3x}) = \frac{6\sqrt{3x}}{(\sqrt{3x})^2 - 9} = \frac{6\sqrt{3x}}{3x - 9}.$$

$$\begin{aligned} 3x \geq 0 \Rightarrow x \geq 0, \qquad & \sqrt{3x} \in D_f \\ x \geq 0, \qquad & \sqrt{3x} \neq \pm 3 \\ x \geq 0, \qquad & \sqrt{3x} \neq 3 \\ x \geq 0, \qquad & 3x \neq 9 \\ x \geq 0, \qquad & x \neq 3 \end{aligned}$$

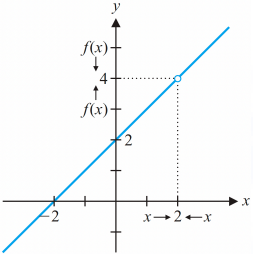
$$\therefore \operatorname{Dom}(f \circ g) = [0, \infty) \setminus \{3\}.$$

Limit

Illustration : One sided limit

- The function $f(x) = \dfrac{x^2-4}{x-2}$ is undefined at $x = 2$, i.e $f(2)$ does not exist.
- Can we have some information about $f(x)$ around the point $x = 2$?
- We compute some values of the function $f(x)$ when x is close to 2

x	$\dfrac{x^2-4}{x-2}$
1.9	3.9
1.99	3.99
1.999	3.999
1.9999	3.9999



x	$\dfrac{x^2-4}{x-2}$
2.1	4.1
2.01	4.01
2.001	4.001
2.0001	4.0001

- From the first table, as $x \to 2$ from the left ($x < 2$), then $f(x) \to 4$. Formally, we write this as
$$\lim_{x \to 2^-} \frac{x^2-4}{x-2} = 4.$$
- From the second table, as $x \to 2$ from the right ($x > 2$), then $f(x) \to 4$. Formally write this as
$$\lim_{x \to 2^+} \frac{x^2-4}{x-2} = 4$$

Illustration : The existence of limit

- We call $\lim_{x\to 2^-} f(x)$ & $\lim_{x\to 2^+} f(x)$ **one-sided limits**.
- Since we have

$$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x) = 4$$

then we conclude

$$\lim_{x\to 2} f(x) = 4$$

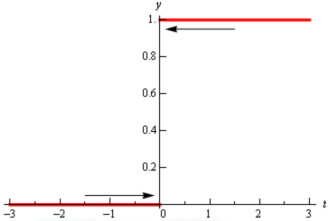
- We can also determine this limit algebraically:

$$\begin{aligned}
\lim_{x\to 2} f(x) &= \lim_{x\to 2} \frac{x^2-4}{x-2} \\
&= \lim_{x\to 2} \frac{(x-2)(x+2)}{x-2} && \text{(Cancel the factors of } (x-2)\text{)} \\
&= \lim_{x\to 2} (x+2) && \text{(As } x \text{ approaches } 2\text{, } (x+2) \text{ approaches } 4.\text{)} \\
&= 4.
\end{aligned}$$

One-sided limit : A limit that doesn't exist

Example 5. Estimate $\lim_{t\to 0} H(t)$ where $H(t)$ is the Heaviside step function given by

$$H(t) = \begin{cases} 0, & t < 0, \\ 1, & t \geq 0. \end{cases}$$



- As t approaches 0 from the right $(t > 0)$, the values of $H(t)$ move toward 1.

$$\lim_{t\to 0^+} H(t) = 1$$

- As t approaches 0 from the left $(t < 0)$, the values of $H(t)$ move toward 0.

$$\lim_{t\to 0^-} H(t) = 0$$

$\Rightarrow$ Since the left and right limits are not equal,

$$\boxed{\lim_{t\to 0} H(t) \text{ does not exist.}}$$

One-Sided and Two-Sided Limits

Right-handed limit

$$\lim_{x \to a^+} f(x) = L$$

when $f(x)$ gets close to L as x approaches a **from the right** ($x > a$, $x \neq a$).

Left-handed limit

$$\lim_{x \to a^-} f(x) = L$$

when $f(x)$ gets close to L as x approaches a **from the left** ($x < a$, $x \neq a$).

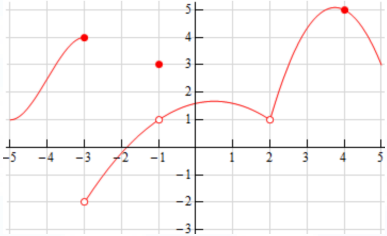
Existence of the limit

$$\boxed{\lim_{x \to a} f(x) = L \quad \text{if and only if} \quad \lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L.}$$

If these one-sided limits differ or do not exist, the **limit does not exist**.

One sided Limit: determining limit from graph

Example 6. For the graph of $f(x)$ below, determine $f(a)$ and $\lim_{x\to a} f(x)$ for each given value of a: $a = -3, -1, 2, 4$.



(a) $a = -3$

- From the graph:

$$f(-3) = 4$$

$$\lim_{x\to -3^-} f(x) = 4$$

$$\lim_{x\to -3^+} f(x) = -2$$

$\Rightarrow$ Since the one-sided limits differ,

(b) $a = -1$

- From the graph:

$$f(-1) = 3$$

$$\lim_{x\to -1^-} f(x) = 1$$

$$\lim_{x\to -1^-} f(x) = 1$$

$\Rightarrow$ The limit exists

Computing limits: Limit properties

Limit Properties

Suppose $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ exist and c is a constant. Then:

1. $\lim_{x\to a} c = c$.
2. $\lim_{x\to a} x = a$.
3. $\lim_{x\to a} [c\, f(x)] = c \lim_{x\to a} f(x)$.
4. $\lim_{x\to a} (f(x) \pm g(x)) = \lim_{x\to a} f(x) \pm \lim_{x\to a} g(x)$.
5. $\lim_{x\to a} (f(x)\, g(x)) = (\lim_{x\to a} f(x))(\lim_{x\to a} g(x))$.
6. (if $\lim_{x\to a} g(x) \neq 0$): $\displaystyle\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$.
7. $\lim_{x\to a} \left(f(x)\right)^n = L^n$, for any $n \in \mathbb{R}$ where L^n is defined.
8. $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{L}$, for any integer $n > 0$ (if n even, assume $L \geq 0$).
9. $\lim_{x\to a} c^x = c^a$

Example : Limit of rational function (direct)

Example 7. Calculate

$$\lim_{z\to 1} \frac{6-3z+10z^2}{-2z^4+7z^3+1}$$

Solution

$$\lim_{z\to 1} \frac{6-3z+10z^2}{-2z^4+7z^3+1} = \frac{\lim_{z\to 1}(6-3z+10z^2)}{\lim_{z\to 1}(-2z^4+7z^3+1)} \quad \text{(Quotient rule; need denom. limit } \neq 0\text{)}$$

$$= \frac{6-3[(1)]+10(1)^2}{-2(1)^4+7(1)^3+1} \quad \text{(direct substitution for polynomials)}$$

$$= \frac{13}{6}.$$

Example : Limit of rational function (by factoring)

If we encounter $\frac{0}{0}$ while substituting, we need to work on the function a bit more.

Example 8. Find the limit of

$$\lim_{x\to 2} \frac{x^2+4x-12}{x^2-2x}$$

Solution.

$$\lim_{x\to 2} \frac{x^2+4x-12}{x^2-2x} \neq \frac{\lim_{x\to 2}(x^2+4x-12)}{\lim_{x\to 2}(x^2-2x)}$$ Denominator limit $= 0$; quotient rule not applicable

$$\lim_{x\to 2} \frac{x^2+4x-12}{x^2-2x} = \lim_{x\to 2} \frac{(x-2)(x+6)}{x(x-2)}$$ Factor numerator and denominator

$$= \lim_{x\to 2} \frac{x+6}{x}$$ Cancel $(x-2)$; valid since $x \to 2$ but $x \neq 2$

$$= \frac{2+6}{2}$$ (direct substitution)

$$= 4.$$

Example: Finding a Limit by Rationalizing

Example 9. Evaluate $\lim_{x\to 0} \frac{\sqrt{x+2}-\sqrt{2}}{x}$ **Solution.**

$$\lim_{x\to 0} \frac{\sqrt{x+2}-\sqrt{2}}{x} = \lim_{x\to 0} \frac{(\sqrt{x+2}-\sqrt{2})}{x} \frac{(\sqrt{x+2}+\sqrt{2})}{(\sqrt{x+2}+\sqrt{2})}$$ Multiply by conjugate

$$= \lim_{x\to 0} \frac{(x+2)-2}{x(\sqrt{x+2}+\sqrt{2})}$$ use $(a-b)(a+b) = a^2 - b^2$

$$= \lim_{x\to 0} \frac{x}{x(\sqrt{x+2}+\sqrt{2})}$$

$$= \lim_{x\to 0} \frac{1}{\sqrt{x+2}+\sqrt{2}}$$ Cancel x (valid since $x \neq 0$)

$$= \frac{1}{\sqrt{2}+\sqrt{2}}$$ substitute $x = 2$

$$= \frac{1}{2\sqrt{2}}.$$

Limit of trigonometry functions

Theorem 2 (Limits of Trigonometric Functions)

For every real number c in the function's domain:

1. $\lim_{t\to c} \sin t = \sin c$
2. $\lim_{t\to c} \cos t = \cos c$
3. $\lim_{t\to c} \tan t = \tan c$
4. $\lim_{t\to c} \cot t = \cot c$
5. $\lim_{t\to c} \sec t = \sec c$
6. $\lim_{t\to c} \csc t = \csc c$

Example 10. Find $\lim_{t\to 0} \dfrac{t^2 \cos t}{t+1}$.

Solution. Using product rules and direct substitution we get

$$\lim_{t\to 0} \frac{t^2 \cos t}{t+1} = \left(\lim_{t\to 0} \frac{t^2}{t+1}\right)\left(\lim_{t\to 0} \cos t\right) = \frac{0}{1} \cdot 1 = 0.$$

Special Trigonometric Limits

(1) $\lim_{t\to 0} \dfrac{\sin t}{t} = 1$ (2) $\lim_{t\to 0} \dfrac{1-\cos t}{t} = 0$

Example : Special Trigonometry limits

Example 11. Find each limit

(a) $\lim_{x\to 0} \frac{\sin(3x)}{x}$ (b) $\lim_{t\to 0} \frac{1-\cos t}{\sin t}$ (c) $\lim_{x\to 0} \frac{\sin(4x)}{\tan x}$

Solution. (a) Multiplying and dividing by 3 we get

$$\lim_{x\to 0} \frac{\sin(3x)}{x} = \lim_{x\to 0} 3\,\frac{\sin(3x)}{3x} = 3\lim_{x\to 0} \frac{\sin(3x)}{3x} = 3(1) = 3.$$

(b) We multiply and divide by t to get

$$\lim_{t\to 0} \frac{1-\cos t}{\sin t} = \lim_{t\to 0} \left(\frac{1-\cos t}{t}\right)\left(\frac{t}{\sin t}\right) = \left(\lim_{t\to 0} \frac{1-\cos t}{t}\right)\left(\lim_{t\to 0} \frac{t}{\sin t}\right) = 0\cdot 1 = 0.$$

(c) Here, we use the fact that $\tan x = \frac{\sin x}{\cos x}$

$$\lim_{x\to 0} \frac{\sin(4x)}{\tan x} = \lim_{x\to 0} \frac{\sin(4x)}{\sin x}\cos x = 4\left(\lim_{x\to 0} \frac{\sin(4x)}{4x}\right)\left(\lim_{x\to 0} \frac{x}{\sin x}\right)\left(\lim_{x\to 0} \cos x\right)$$

$$= 4(1)(1)(1) = 4.$$

Theorem 3 (Squeeze Theorem)

Suppose that for all x in $[a, b]$ (except possibly at $x = c$), we have

$$f(x) \leq h(x) \leq g(x).$$

If

$$\lim_{x \to c} f(x) = \lim_{x \to c} g(x) = L,$$

for some $a \leq c \leq b$. Then,

$$\lim_{x \to c} h(x) = L.$$

Example 12. Evaluate $\lim_{x \to 0} x^2 \cos\left(\frac{1}{x}\right)$.

Solution. We note that for $x \neq 0$

$$-1 \leq \cos\left(\frac{1}{x}\right) \leq 1 \qquad \text{Basic property of cosine}$$

$$-x^2 \leq x^2 \cos\left(\frac{1}{x}\right) \leq x^2 \qquad \text{Multiply through by } x^2 > 0$$

Since $\lim_{x \to 0}(-x^2) = 0$ & $\lim_{x \to 0}(x^2) = 0$ then by the Squeeze Theorem

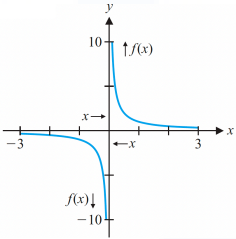
$$\lim_{x \to 0} x^2 \cos\left(\frac{1}{x}\right) = 0$$

Limit involving infinities

Example 13. Examine $\lim_{x\to 0} \frac{1}{x}$.

- The function $f(x) = \frac{1}{x}$ is undefined at $x = 0$, but we can study its one side limit. We compute several values of $f(x)$ when x is close to 0.

x	$f(x) = \frac{1}{x}$
0.1	10
0.01	100
0.001	1000
0.0001	10000



x	$f(x) = \frac{1}{x}$
-0.1	-10
-0.01	-100
-0.001	-1000
-0.0001	-10000

- As $x \to 0^+$, $\frac{1}{x}$ increases without bound.

$$\lim_{x\to 0^+} \frac{1}{x} = +\infty$$

- As $x \to 0^-$, $\frac{1}{x}$ decreases without bound.

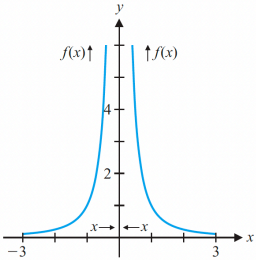
$$\lim_{x\to 0^-} \frac{1}{x} = -\infty$$

Infinite Limit (Both Sides)

Example 14. Examine $\lim_{x\to 0} \frac{1}{x^2}$.

- The function $f(x) = \frac{1}{x^2}$ is undefined at $x = 0$. We study its one-sided behavior by computing values for x close to 0.

x	$f(x) = \frac{1}{x^2}$
0.1	100
0.01	$10{,}000$
0.001	10^6
0.0001	10^8



x	$f(x) = \frac{1}{x^2}$
-0.1	100
-0.01	$10{,}000$
-0.001	10^6
-0.0001	10^8

- As $x \to 0^+$ and $x \to 0^-$, $\frac{1}{x^2}$ increases without bound. As $x \to 0^-$. So

$$\lim_{x\to 0^+} \frac{1}{x^2} = +\infty, \qquad \lim_{x\to 0^-} \frac{1}{x^2} = +\infty$$

- Since both one-sided limits agree (both $+\infty$), we write

$$\lim_{x\to 0} \frac{1}{x^2} = +\infty,$$

Vertical Asymptote

Intuitive Idea

- A **vertical asymptote** is a vertical line that the graph of a function gets closer and closer to, but never touches.
- As x gets near a certain value a, the function $f(x)$ becomes very large (goes to $+\infty$) or very small (goes to $-\infty$).

Formal Definition

The function $f(x)$ has a vertical asymptote at $x = a$ if any of the following limits hold:

$$\lim_{x \to a} f(x) = \pm\infty,$$

$$\lim_{x \to a^+} f(x) = \pm\infty,$$

$$\lim_{x \to a^-} f(x) = \pm\infty.$$

In this case, we say the line $x = a$ is a **vertical asymptote** of the graph of $f(x)$.

Infinite One-Sided Limits Disagree

Example 15. Evaluate $\lim_{x\to 3^+} \frac{2x}{x-3}$, $\lim_{x\to 3^-} \frac{2x}{x-3}$, $\lim_{x\to 3} \frac{2x}{x-3}$.

$$\lim_{x\to 3^+} \frac{2x}{x-3} = \lim_{x\to 3^+} \frac{\overbrace{2x}^{\to 6 > 0}}{\underbrace{x-3}_{\to 0^+}}$$

pos. constant over tiny pos. $\Rightarrow +\infty$

$$= +\infty.$$

$$\lim_{x\to 3^-} \frac{2x}{x-3} = \lim_{x\to 3^-} \frac{\overbrace{2x}^{\to 6 > 0}}{\underbrace{x-3}_{\to 0^-}}$$

pos. constant over tiny neg. $\Rightarrow -\infty$

$$= -\infty.$$

$\therefore \lim_{x\to 3} \frac{2x}{x-3}$ does not exist

one-sided limits are unequal.

Conclusion: $x = 3$ *is a vertical asymptote.*

Limits at infinity

Limits at Infinity & Horizontal Asymptotes

- We are also interested to see how $f(x)$ behaves as x grows without bound ($x \to \infty$) or decreases without bound ($x \to -\infty$)
- For $f(x) = \dfrac{1}{x}$ we observe that $f(x) \to 0$ as $|x| \to \infty$:

$$\lim_{x\to\infty} \frac{1}{x} = 0 \qquad \text{and} \qquad \lim_{x\to-\infty} \frac{1}{x} = 0.$$

Horizontal Asymptote (definition)

The line $y = L$ is a **horizontal asymptote** of f if

$$\lim_{x\to\infty} f(x) = L \quad \text{or} \quad \lim_{x\to-\infty} f(x) = L.$$

For $f(x) = \dfrac{1}{x}$, the horizontal asymptote is $y = 0$.

Example: Finding a Horizontal Asymptote

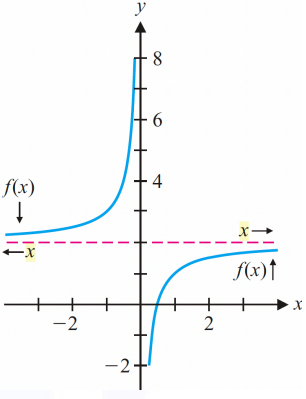
Example 16. Find any horizontal asymptotes of the graph of $f(x) = 2 - \dfrac{1}{x}$.

- As $x \to \infty$, we have $\dfrac{1}{x} \to 0$, hence

$$\lim_{x\to\infty}\left(2 - \frac{1}{x}\right) = 2.$$

- Similarly, as $x \to -\infty$,

$$\lim_{x\to-\infty}\left(2 - \frac{1}{x}\right) = 2.$$



Conclusion: Since both one-sided limits at infinity are 2, the line $y = 2$ is a **horizontal asymptote**.

Example 17. Evaluate (a) $\lim_{x\to\infty}(2x^4 - x^2 - 8x)$ and (b) $\lim_{t\to-\infty}\left(\frac{1}{3}t^5 + 2t^3 - t^2 + 8\right)$.

(a) $\lim_{x\to\infty}(2x^4 - x^2 - 8x) = \infty - \infty - \infty$ (naive term-by-term; indeterminate)

$$\lim_{x\to\infty}(2x^4 - x^2 - 8x) = \lim_{x\to\infty}\left[x^4\left(2 - \frac{1}{x^2} - \frac{8}{x^3}\right)\right] \qquad \text{factor \underline{largest power}: } x^4$$

$$= \infty \cdot 2 = \infty.$$

(b) Again, naive substitution term by term will give us indeterminate form $\infty - \infty$. Proceeding as before we have

$$\lim_{t\to-\infty}\left(\frac{1}{3}t^5 + 2t^3 - t^2 + 8\right) = \lim_{t\to-\infty}\left[t^5\left(\frac{1}{3} + \frac{2}{t^2} - \frac{1}{t^3} + \frac{8}{t^5}\right)\right] \qquad \text{factor \underline{largest power} } t^5$$

$$= (-\infty)\cdot\frac{1}{3} = -\infty.$$

Observation: leading–term rule

If $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ is a polynomial of degree n (with $a_n \neq 0$), then

$$\lim_{x\to\infty} p(x) = \lim_{x\to\infty} a_n x^n, \quad \& \quad \lim_{x\to-\infty} p(x) = \lim_{x\to-\infty} a_n x^n.$$

Intuition: as $|x|$ grows, the highest power term $a_n x^n$ dominates all others.

Example 18. Evaluate $\lim_{x\to\infty} \dfrac{2x^4 - x^2 + 8x}{-5x^4 + 7}$ and $\lim_{x\to-\infty} \dfrac{2x^4 - x^2 + 8x}{-5x^4 + 7}$.

Part a. $\lim_{x\to\infty} \frac{2x^4-x^2+8x}{-5x^4+7} = \frac{\infty}{-\infty}$

$$\lim_{x\to\infty} \frac{2x^4 - x^2 + 8x}{-5x^4 + 7} = \lim_{x\to\infty} \frac{x^4\left(2 - \frac{1}{x^2} + \frac{8}{x^3}\right)}{x^4\left(-5 + \frac{7}{x^4}\right)}$$ factor x^4 from top/bottom

$$= \lim_{x\to\infty} \frac{2 - \frac{1}{x^2} + \frac{8}{x^3}}{-5 + \frac{7}{x^4}}$$ cancel x^4 $(x \neq 0)$

$$= \frac{2 + 0 + 0}{-5 + 0}$$ $\frac{1}{x^k} \to 0$ for $k > 0$

$$= -\frac{2}{5}.$$

Part b.

$$\lim_{x\to-\infty} \frac{2x^4 - x^2 + 8x}{-5x^4 + 7} = \lim_{x\to-\infty} \frac{2 - \frac{1}{x^2} + \frac{8}{x^3}}{-5 + \frac{7}{x^4}}$$ same leading-power factor/cancel

$$= \frac{2}{-5}$$ even power $\Rightarrow$ signs unchanged $= -\frac{2}{5}$

Example 19. Evaluate (a) $\lim_{x\to\infty} \frac{\sqrt{3x^2+6}}{5-2x}$ and (b) $\lim_{x\to-\infty} \frac{\sqrt{3x^2+6}}{5-2x}$.

First we observe that

$$\frac{\sqrt{3x^2+6}}{5-2x} = \frac{\sqrt{x^2\left(3+\frac{6}{x^2}\right)}}{x\left(\frac{5}{x}-2\right)}$$

factor x^2 inside root; factor x below

$$= \frac{|x|\sqrt{3+\frac{6}{x^2}}}{x\left(\frac{5}{x}-2\right)}$$

$\sqrt{x^2} = |x|$

(a) When $x \to \infty$, then we are interested in $x > 0$ so $|x| = x$. Hence

$$\lim_{x\to\infty} \frac{\sqrt{3x^2+6}}{5-2x} = \lim_{x\to\infty} \frac{x\sqrt{3+\frac{6}{x^2}}}{x\left(\frac{5}{x}-2\right)} = \lim_{x\to\infty} \frac{\sqrt{3+\frac{6}{x^2}}}{\frac{5}{x}-2} = \frac{\sqrt{3+0}}{0-2} = -\frac{\sqrt{3}}{2}.$$

(b) When $x \to -\infty$, then we are interested in $x < 0$ so $|x| = -x$. Hence

$$\lim_{x\to-\infty} \frac{\sqrt{3x^2+6}}{5-2x} = \lim_{x\to-\infty} \frac{-x\sqrt{3+\frac{6}{x^2}}}{x\left(\frac{5}{x}-2\right)} = \lim_{x\to-\infty} \frac{-\sqrt{3+\frac{6}{x^2}}}{\frac{5}{x}-2} = \frac{-\sqrt{3+0}}{0-2} = \frac{\sqrt{3}}{2}.$$

Continuity

Definition

- A function $f(x)$ is said to be **continuous at** $x = a$ if
$$\lim_{x \to a} f(x) = f(a)$$
- A function is said to be continuous on the interval $[a, b]$ if it is continuous at each point in the interval.

Note: This definition implicitly assumes that both $f(a)$ and $\lim_{x \to a} f(x)$ exist. If either does not exist, the function will not be continuous at $x = a$.

From the definition, to assert that a function is continuous, we need to check the following

Fact

$f(x)$ is continuous at $x = a$, if and only if

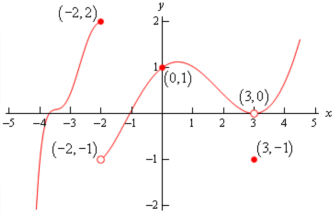
(i) $\lim_{x \to a^-} f(x) = L$

(ii) $\lim_{x \to a^+} f(x) = L$

(iii) $L = f(a)$

Example : Identifying continuous function from the graph

Example 20. Given the graph of $f(x)$, shown below, determine if $f(x)$ is continuous at $x = -2, 0$ and $x = 3$



Solution.

- At $x = -2$

$$f(-2) = 2, \qquad \lim_{x \to -2} f(x) \text{ does not exist}$$

Not continuous at $x = -2$ (one–sided limits disagree).

- At $x = 0$

$$f(0) = 1, \qquad \lim_{x \to 0} f(x) = 1$$

Continuous at $x = 0$ (value equals the limit).

- At $x = 3$

$$f(3) = -1, \qquad \lim_{x \to 3} f(x) = 0$$

Not continuous at $x = 3$(Function value disagree with the limit)

Continuity $\Longrightarrow$ "commutes with limits"

If f is continuous at c, then

$$\lim_{x\to c} f(x) \;=\; f(c) \;=\; f\Big(\lim_{x\to c} x\Big).$$

Composition rule

If $f(x)$ is continuous at $x = b$ and $\lim_{x\to a} g(x) = b$, then

$$\lim_{x\to a} f(g(x)) \;=\; f\Big(\lim_{x\to a} g(x)\Big).$$

Example 21. Evaluate $\lim_{x\to 0} e^{\sin x}$.

Since the exponential e^u is continuous for all u and $\lim_{x\to 0} \sin x = 0$,

$$\lim_{x\to 0} e^{\sin x} \;=\; e^{\lim_{x\to 0} \sin x} \;=\; e^0 \;=\; 1.$$

Intermediate Value Theorem

Continuity of a Function on an Interval

Consider the following situation:

- Suppose f is continuous on the interval $[a, b]$, with $f(a) < 0$ and $f(b) > 0$.
- Is it possible that the graph of $f(x)$ does not intersect the x-axis ($y = 0$)?
- No. Since f is continuous, its graph must cross the x-axis at least once, at some point $x = c$.

- This means that if f is continuous on an interval $[a, b]$ and

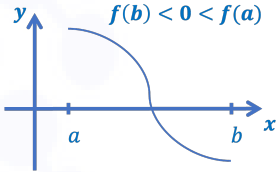
$$f(a) < 0 < f(b)$$

or

$$f(b) < 0 < f(a),$$

then there exists at least one point $c \in (a, b)$ such that

$$f(c) = 0.$$



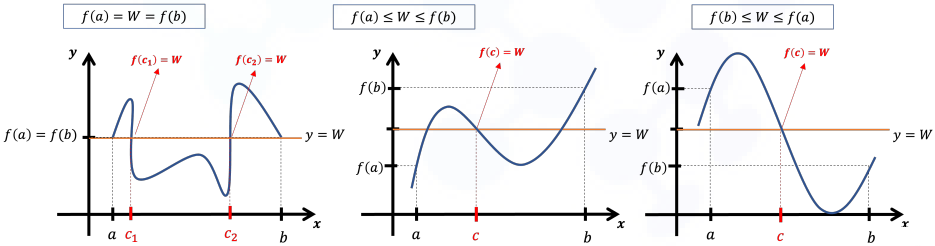
Theorem 4 (Intermediate Value Theorem)

Let f be continuous on the closed interval $[a, b]$. If W is any number between $f(a)$ and $f(b)$, that is

$$f(a) \leq W \leq f(b) \quad \text{or} \quad f(b) \leq W \leq f(a),$$

then there exists at least one point $c \in [a, b]$ such that

$$f(c) = W.$$



Theorem 5 (Intermediate Value Theorem)

Let f be continuous on $[a, b]$. If W is any numberbetween $f(a)$ and $f(b)$, that is

$$f(a) \leq W \leq f(b) \quad \textit{or} \quad f(b) \leq W \leq f(a).$$

Then there exists $c \in [a, b]$ such that

$$f(c) = W.$$

Remarks (How to Apply the Theorem):

- Identify the function f, the interval $[a, b]$, and the value W.
- Verify that W lies between $f(a)$ and $f(b)$:

  $$f(a) \leq W \leq f(b) \quad \text{or} \quad f(b) \leq W \leq f(a).$$

  another way to write it is $W \in f([a, b])$
- Make sure that f is continuous on $[a, b]$.

Remark on existence: The theorem guarantees the **existence** of such a point c, but it not tell us exactly which c it is.

Example 22. Use the Intermediate Value Theorem to show that

$$x - \cos x = 0$$

has a solution between $x = 0$ and $x = \frac{\pi}{2}$.

Solution. Step 1. Identify f, $[a, b]$, and W Let

$$f(x) = x - \cos x, \qquad [a, b] = \left[0, \frac{\pi}{2}\right], \qquad W = 0$$

Step 2. We show that f is continuous on $[0, \frac{\pi}{2}]$. Since both x and $\cos x$ are continuous on $\mathbb{R}$, then the function $f(x) = x - \cos x$ is continuous on all $\mathbb{R}$, particularly on $[0, \frac{\pi}{2}]$.

Step 3. Ensure that $f(a) \leq W \leq f(b)$ or $f(b) \leq W \leq f(a)$. Compute:

$$\begin{aligned} f(0) &= 0 - \cos 0 = -1 < \mathbf{0}, \\ f\left(\tfrac{\pi}{2}\right) &= \tfrac{\pi}{2} - \cos\left(\tfrac{\pi}{2}\right) = \tfrac{\pi}{2} > \mathbf{0}. \end{aligned} \qquad \Longrightarrow \qquad f(0) < \mathbf{0} < f\left(\frac{\pi}{2}\right),$$

Conclusion: Since $f(x) = x - \cos x$ is continuous on $[0, \frac{\pi}{2}]$ and $f(0) < 0 < f\left(\frac{\pi}{2}\right)$, by the Intermediate Value Theorem there exists $c \in [0, \frac{\pi}{2}]$ such that

$$f(c) = 0 \quad \Longleftrightarrow \quad c - \cos c = 0.$$

Example

Example 23. Use the Intermediate Value Theorem to show that

$$x - \cos x = 0$$

has a solution between $x = 0$ and $x = \frac{\pi}{2}$. Can we obtain a better approximation of the solution?

Solution.
Can we obtain a better approximation of the solution? That is, can we determine a smaller interval than $[0, \frac{\pi}{2}]$ that still contains the solution? Use the Bisection method.

Consider the midpoint of the interval $[0, \frac{\pi}{2}]$, namely $\frac{\pi}{4}$. When computing $f(\frac{\pi}{4})$ we obtain

$$f\left(\frac{\pi}{4}\right) = \frac{\pi}{4} - \cos\left(\frac{\pi}{4}\right) = \frac{\pi}{4} - \frac{\sqrt{2}}{2} > 0.$$

Since $f(x) = x - \cos x$ is continuous on $[0, \frac{\pi}{4}]$ and

$$f(0) < 0 < f\left(\frac{\pi}{4}\right),$$

by the Intermediate Value Theorem there exists $c \in [0, \frac{\pi}{4}]$ such that

$$f(c) = 0 \quad \Longleftrightarrow \quad c - \cos c = 0.$$

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